

System Identification Method for Robotic Manipulator based on Dynamic Momentum Regressor

Kyongho Park and Youngjin Choi

Abstract—The paper presents a momentum regressor using both numerical integration and Hamiltonian dynamic description for robotic manipulator. It has an advantage in that the proposed momentum regressor does not require the acceleration estimation or sensor for the identification of dynamic parameters. In addition, the identification algorithm is proposed by using Lagrange multiplier and bordered gramian matrix for the minimization of difference between the previous and the current estimates of dynamic parameters. The developed algorithm is intuitive and easy to implement in real-time fashion. Finally, the effectiveness of the proposed momentum regressor and identification method is shown through the numerical simulations about a planar two-link robotic manipulator system.

I. INTRODUCTION

For precise control of robotic manipulators, exact knowledge about robotic motion description including its dynamic parameters such as link inertias, masses and friction coefficients are required. For this purpose, an appropriate identification algorithm is also needed because various friction coefficients are mainly varying according to the assembling and manufacturing procedures, even though the dynamic parameters are considered from the initial design phase. A variety of identification schemes for linear and nonlinear systems have suggested in [1]. In addition, since the identification on the dynamic parameters of robotic manipulator has been one of the classical research topics, several papers and textbooks have dealt with the topic [2], [3], [4], [5]. In the paper, novel identification method is proposed by using new form of regressor, Lagrange multiplier and bordered gramian matrix through the minimization of difference between the previous and the current estimations of dynamic parameters. In the previous researches, the actual acceleration information was replaced into the desired acceleration one thanks to the adaptive control scheme. However, the paper does not contain the desired trajectory and the adaptive control scheme, but focuses only on the identification itself.

Since the identification of dynamic parameters for robotic manipulator is basically based on the classical mechanics (equations of motion), let us review well-known three kinds of dynamic descriptions such as Newton-Euler equation, Euler-Lagrange equation, and the principle of virtual work.

K. Park is with the School of Mechanical Engineering (also he belongs to the LS Mtron Co.), POSTECH, 37673, South Korea. e-mail: neotele@lsmtron.com

Y. Choi is with the School of Electrical Engineering, Hanyang University, 15588, South Korea. e-mail: cyj@hanyang.ac.kr

This work was supported in part by the Convergence Technology Development Program for Bionic Arm through the National Research Foundation of Korea funded by the Ministry of Science, ICT & Future Planning (NRF-2015M3C1B2052811), and in part by the National Research Foundation of Korea (NRF-2013R1A1A2010192), Republic of Korea.

Then the second-order differential equations are obtained and the algorithms come down to calculating and integrating the acceleration [6]. Actually, aforementioned three methods result in the minimal representation for the corresponding equation of motion if there are not non-holonomic constraints. Generally, expressing the equations of motion with a minimal set of coordinates results in less differential equations, which are, however, more coupled and usually exhibit stronger nonlinearities compared to the non-minimal formulations [7]. The representative non-minimal formulation is the Hamiltonian equation of motion. Its advantage is that it brings a set of first-order differential equations, though it results in the mixed form of differential equations. Actually the Hamiltonian equation of motion is a dual form of the Euler-Lagrange ones because the Legendre transform could be always found due to the positiveness of inertia matrix. In order to define a momentum regressor as new form, Hamiltonian equation of motion will be utilized for further mathematical derivations.

The simulation and control of robotic manipulators demand the knowledge on the values of dynamic parameters. Computing such parameters from the design data of the mechanical structure is not simple, as an alternative, we may adopt CAD (computer aided design) tools which allow computing the values of dynamic parameters (links, actuators, and transmissions) on the basis of their geometry and the type of material employed. Nevertheless, the estimates obtained by using such techniques are inaccurate because of the simplification typically introduced by geometric modeling. Moreover, the complex dynamic effects, such as a variety of joint frictions, may not be taken into account. Thus the robust identification algorithm for the dynamic parameters is required. In order to identify the dynamic parameters of robot manipulator, first, we establish the regressor from the corresponding equation of motion. In [2], the acceleration-based regressor was derived from the conventional Newton-Euler algorithm. In this paper, the momentum regressor is newly defined by using the Hamiltonian equation motion, and it does not require the acceleration measurement or estimation for robotic manipulator.

This paper is organized as follows: section II derives momentum regressor from the classical robot dynamics, section III suggests the real-time identification method by using the optimization subject to the equality constraint, the numerical simulations are shown in section IV, and then we draw the conclusions in section V.

II. MOMENTUM REGRESSOR

The robotic manipulator is generally described using Euler-Lagrange equation of motion as following form:

$$\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} \quad (1)$$

where $q = [q_1, q_2, \dots, q_n]^T$ is an n -dimensional configuration (or joint displacement) vector, $\dot{q}$ a joint velocity vector, $\ddot{q}$ a joint acceleration vector, $M(q)$ an inertia matrix, $C(q, \dot{q})$ a Coriolis and centrifugal matrix, $g(q)$ a gravity vector, $\tau_f(q, \dot{q})$ a frictional torque vector, $J(q)$ a manipulator Jacobian, f_{ext} an external force such as load, and τ a control input torque vector. The natural choice of a model for identification of dynamic parameters is above Euler-Lagrange model of the manipulator, which has the property of linearity in dynamic parameters as following form:

$$\tau = Y_L(q, \dot{q}, \ddot{q})\theta, \quad (2)$$

where $Y_L(q, \dot{q}, \ddot{q})$ is a conventional regressor matrix containing all the nonlinear terms related to the displacement, velocity and acceleration, and θ is a constant dynamic parameter vector such as link inertia, link mass and positions of center of mass, and friction coefficients. The general method constructing the conventional regressor has been suggested in [2], [3]. It is referred to as a Lagrange regressor or conventional regressor. If the control input torque, displacement, velocity, acceleration vectors are all measurable every time instants, then it might be possible to identify the dynamic parameters θ of robotic manipulator, but the acceleration information should be provided for the algorithm implementation.

Since the robotic manipulator system is not normally equipped with the accelerometer, we need to construct the estimator to make acceleration signal from the velocity or displacement through numerical differentiation. It might bring wrong identification results. On the other hand, even though it has an accelerometer, it is not easy to combine different information obtained from two or more sensors because the measurement bandwidths of the sensors are different each other. Thus many previous simulations and experiments might bring inexact identification results. As an alternative, new form of regressor that does not require the numerical differentiation or acceleration information is proposed in the following.

Let us define the momentum vector as follow:

$$p = M(q)\dot{q}, \quad (3)$$

where p is an n -dimensional momentum vector derived from the product of inertia matrix and joint velocity vector. Also the above is referred to as a Legendre transformation and it is always possible if and only if the inertia matrix is positive definite and symmetric. In addition, we know that the time derivative of the inertia matrix has the following relation with Coriolis and centrifugal matrix [3]:

$$\dot{M}(q) = C(q, \dot{q}) + C^T(q, \dot{q}), \quad (4)$$

where it can be interpreted that the time derivative of inertia matrix is equal to two times of even part of Coriolis and centrifugal matrix.

After taking a time derivative of Eq. (3), if we rearrange it by using Eqs. (1) and (4), then we have

$$\begin{aligned} \dot{p} &= \dot{M}(q)\dot{q} + M\ddot{q} \\ &= [C(q, \dot{q}) + C^T(q, \dot{q})]\dot{q} \\ &\quad - [C(q, \dot{q})\dot{q} + g(q) + \tau_f(q, \dot{q}) + J^T(q)f_{ext} - \tau] \\ &= C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext} + \tau, \end{aligned} \quad (5)$$

where the above implies the Hamiltonian equation of motion that does not explicitly include an acceleration term. This is the reason why we utilize the Hamiltonian representation for equation of motion. Now let us define first subregressor as

$$\dot{p} = W_1(q, \dot{q})\theta + \tau, \quad (6)$$

where

$$W_1(q, \dot{q})\theta \triangleq C^T(q, \dot{q})\dot{q} - g(q) - \tau_f(q, \dot{q}) - J^T(q)f_{ext}, \quad (7)$$

in which note that the first subregressor is not dependent on the acceleration. Also second subregressor can be defined from the momentum equation as:

$$\begin{aligned} p &= M(q)\dot{q} \\ &\triangleq W_2(q, \dot{q})\theta. \end{aligned} \quad (8)$$

After integrating both sides of Eq. (6), if Eq. (8) is applied, then the new form of regressor is obtained as:

$$u(t) = Y_H(q, \dot{q}, t)\theta, \quad (9)$$

where

$$u(t) \triangleq \int_0^t \tau \, dt \quad (10)$$

$$Y_H(q, \dot{q}, t) \triangleq W_2(q, \dot{q}) - \int_0^t W_1(q, \dot{q}) \, dt \quad (11)$$

in which $u(t)$ implies the time integration of control input torque from the initial time 0 to the current time t , and $Y_H(q, \dot{q}, t)$ is referred to as “momentum regressor” composed of three independent variables such as q , $\dot{q}$, and time t in the paper. If above momentum regressor is utilized, then we know that it does not require the acceleration information but the numerical integration of the applied input torques and first subregressor. The numerical integration is more stable and exact than the numerical differentiation in viewpoint of noise removal. As a matter of fact, Eq. (9) is equal to the time integration form of conventional Lagrange regressor of Eq. (2).

The following section deals with an identification method obtained by applying the Lagrange multiplier and bordered gramian methods in order to solve the optimization problem.

III. IDENTIFICATION ALGORITHM USING MOMENTUM REGRESSOR

For given joint displacement and velocity, the momentum regressor can be determined as a real-time form. Furthermore, the pairs of the integration of input torque and the momentum regressor, $\{u[i], Y_H[i]\}$ at the i th sample, are sequentially obtained as follows:

$$u[i] = Y_H[i]\theta, \quad (12)$$

where

$$Y_H[i] = W_2[i] - \sum_{k=1}^i W_1[k]\delta t \quad (13)$$

$$u[i] = \sum_{k=1}^i \tau[k]\delta t \quad (14)$$

in which δt implies the sampling time interval, namely, time $t = i \cdot \delta t$. Since the measurements $\{u[i], Y_H[i]\}$ for $i = 1, 2, \dots, n$ contain the information about constant dynamic parameters to be identified, it is natural to choose new estimate vector $\hat{\theta}[i]$ so as to minimize the parameter variation energy satisfying the constraint (12). This is a typical least squares problem in [8]. First let us make a normal equation by multiplying $Y_H^T[i]$ with both sides of Eq. (12) as follow:

$$Y_H^T[i]u[i] = (Y_H^T[i]Y_H[i])\hat{\theta}[i] \quad \text{for } i = 1, 2, \dots \quad (15)$$

Second, since above equality should be satisfied for all time instants, it can be accumulated till the current sample i as follow:

$$\sum_{k=1}^i Y_H^T[k]u[k] = \left(\sum_{k=1}^i Y_H^T[k]Y_H[k] \right) \hat{\theta}[i]. \quad (16)$$

On the other hand, since the parameter vector $\hat{\theta}[i]$ to be identified should arrive at the constant values, it can be formulated as the minimization of parameter variation energy with the equality constraint of Eq. (16) as follows:

$$\begin{aligned} \min \frac{1}{2} \left\| \hat{\theta}[i] - \hat{\theta}[i-1] \right\|^2 \\ \text{subject to } \Gamma[i] = \Omega[i]\hat{\theta}[i] \end{aligned} \quad (17)$$

where

$$\Gamma[i] \triangleq \sum_{k=1}^i Y_H^T[k]u[k] \quad (18)$$

$$\Omega[i] \triangleq \sum_{k=1}^i Y_H^T[k]Y_H[k]. \quad (19)$$

Now if we introduce a Lagrangian multiplier method to handle above minimization, we have

$$\begin{aligned} L = \frac{1}{2} \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right)^T \left(\hat{\theta}[i] - \hat{\theta}[i-1] \right) \\ + \lambda^T \left(\Gamma[i] - \Omega[i]\hat{\theta}[i] \right), \end{aligned} \quad (20)$$

where λ is a Lagrange multiplier vector with appropriate dimension. Taking the derivatives with respect to $\hat{\theta}[i]$ and λ , we have

$$\hat{\theta}[i] - \hat{\theta}[i-1] - \Omega^T[i]\lambda = 0 \quad (21)$$

$$\Gamma[i] - \Omega[i]\hat{\theta}[i] = 0. \quad (22)$$

By rearranging above two equations, we can find the matrix-vector equation

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix} \begin{bmatrix} \hat{\theta}[i] \\ -\lambda \end{bmatrix} = \begin{bmatrix} \hat{\theta}[i-1] \\ \Gamma[i] \end{bmatrix}, \quad (23)$$

where the first matrix is called as ‘‘bordered gramian matrix’’ in [9], [10]. Exact solution of Eq. (23) is given by inverting the bordered gramian matrix. In addition, the concrete procedure to obtain the inverse of the bordered gramian matrix was suggested in [10] and its inverse has the following form:

$$\begin{bmatrix} I & \Omega^T[i] \\ \Omega[i] & 0 \end{bmatrix}^{-1} = \begin{bmatrix} W[i] - W[i]\Omega[i]^T Z[i]\Omega[i]W[i] & W[i]\Omega[i]^T Z[i] \\ Z[i]\Omega[i]^T W[i] & I - Z[i] \end{bmatrix} \quad (24)$$

where

$$W[i] \triangleq (I + \Omega[i]^T \Omega[i])^{-1} \quad (25)$$

$$Z[i] \triangleq (\Omega[i]W[i]\Omega[i]^T)^+, \quad (26)$$

in which the existence of the matrix $W[i]$ is always guaranteed but the matrix $Z[i]$ exists only when $\Omega[i]$ has full rank. Thus we have defined the matrix $Z[i]$ by using the pseudoinverse obtained by using the singular value decomposition, not the inverse. Indeed even though the full rank $Y_H[i]$ is guaranteed, we cannot guarantee full rank $Z[i]$ because the normal equation is an expansion of $Y_H[i]$ towards higher dimension of rectangular $Y_H[i]$. Strictly speaking, the full rank condition of $Z[i]$ is not equal to the persistent excitation (PE) condition in the adaptive control [11], [12], because the PE condition implies just the full rank of $Y_H[i]$ but $Z[i]$ makes use of the summation of normal form of $Y_H[i]$.

Now let us obtain the solution of Eq. (23)

$$\hat{\theta}[i] = \Omega_W^+[i]\Gamma[i] + (I - \Omega_W^+[i]\Omega[i])W[i]\hat{\theta}[i-1] \quad (27)$$

$$\lambda = (I - Z[i])\Gamma[i] + Z[i]\Omega[i]^T W[i]\hat{\theta}[i-1]$$

where

$$\begin{aligned} \Omega_W^+[i] &\triangleq W[i]\Omega[i]^T Z[i] \\ &= W[i]\Omega[i]^T (\Omega[i]W[i]\Omega[i]^T)^+. \end{aligned} \quad (28)$$

Fig. 1 illustrates the proposed identification scheme of Eq. (27). Since the matrix $W[i]$ has the full rank, the following statements are always satisfied in Fig. 1.

- 1) $\mathcal{R}(\Omega_W^+[i])$ and $\mathcal{N}(\Omega[i])$ are $W^{-1}[i]$ weighted orthogonal because

$$\begin{aligned} &(\Omega_W^+[i])^T W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i]W[i]W^{-1}[i](I - \Omega^+[i]\Omega[i]) \\ &= (\Omega[i]W[i]\Omega[i]^T)^+ \Omega[i](I - \Omega^+[i]\Omega[i]) \\ &= 0. \end{aligned}$$

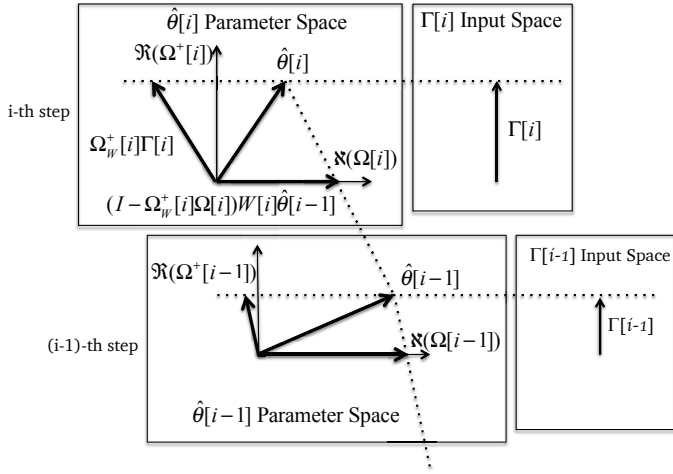


Fig. 1: Geometric interpretation of Eq. (27)

- 2) $\mathcal{R}(\Omega^T[i])$ and $\mathcal{N}(\Omega[i])$ are geometrically orthogonal because

$$(\Omega^T[i])^T (I - \Omega^+[i]\Omega[i]) = \Omega[i](I - \Omega^+[i]\Omega[i]) = 0.$$

We should note that $\mathcal{R}(\Omega^+[i]) = \mathcal{R}(\Omega^T[i])$, $\mathcal{R}(\cdot)$ and $\mathcal{N}(\cdot)$ mean the range space and the null space, respectively. An item 1) implies the orthogonality in the weighted metric and another item 2) implies the geometrical orthogonality between the range space of $\Omega^T[i]$ and null space of $\Omega[i]$.

Since the geometrical orthogonality between $\mathcal{R}(\Omega^+[i])$ and $\mathcal{N}(\Omega[i])$ is satisfied in the parameter space $\hat{\theta}[i]$, we have expressed them as coordinate axes in the orthogonal coordinate system as shown in Fig. 1. The identification algorithm of Eq. (27) is composed of the sum of the component $\Omega_W^+[i]\Gamma[i]$ embedded on $\mathcal{R}(\Omega_W^+[i])$ and another component $(I - \Omega_W^+[i]\Omega[i])W[i]\hat{\theta}[i-1]$ embedded on $\mathcal{N}(\Omega[i])$ which is not geometrically orthogonal to $\mathcal{R}(\Omega_W^+[i])$ but $W^{-1}[i]$ weighted orthogonal to $\mathcal{R}(\Omega_W^+[i])$. The matrix $W^{-1}[i]$ becomes able to rotate the embedded range space with suitable rotation angle, in other words, it changes the orthogonal null space projection into the inclined null space projection efficient for utilization of previous parameter estimate $\hat{\theta}[i-1]$. On the other hand, if $\Omega[i]$ has the full rank, the null space does not exist and the inclined null space projection disappears in the identification algorithm of Eq. (27).

As summary, the Eq. (27) aims at obtaining the optimal estimates of the dynamic parameters by using the mapping $\Omega[i]$ from the input $\Gamma[i]$. During the identification process, if the mapping $\Omega[i]$ loses the rank, then the previous parameter estimates are utilized through the inclined null space projection scheme, ultimately, for improving both identification accuracy and convergence speed. The next section shows the effectiveness of the proposed identification algorithm through several simulations.

IV. SIMULATIONS

Let us assume that a planar two-link manipulator is handling the unknown mass M at the end-effector as shown

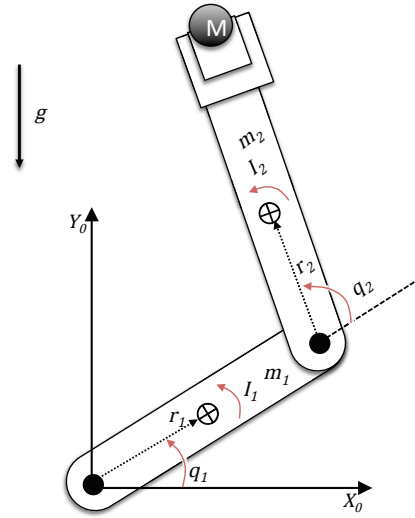


Fig. 2: Typical planar two-link manipulator handling unknown mass M

in Fig. 2, in order for verification of the proposed identification algorithm. The dynamics of Eq. (1) has the following components:

$$M(q) = \begin{bmatrix} I_1 + I_2 + m_2 l_1^2 + 2m_2 r_2 l_1 c_2 & I_2 + m_2 r_2 l_1 c_2 \\ I_2 + m_2 r_2 l_1 c_2 & I_2 \end{bmatrix} \quad (29)$$

$$C(q, \dot{q}) = \begin{bmatrix} -m_2 r_2 l_1 s_2 \dot{q}_2 & -m_2 r_2 l_1 s_2 (\dot{q}_1 + \dot{q}_2) \\ m_2 r_2 l_1 s_2 \dot{q}_1 & 0 \end{bmatrix} \quad (30)$$

$$g(q) = \begin{bmatrix} -m_1 r_1 g c_1 - m_2 l_1 g c_1 - m_2 r_2 g c_{12} \\ -m_2 r_2 g c_{12} \end{bmatrix} \quad (31)$$

$$\tau_f(q, \dot{q}) = \begin{bmatrix} F_{s1} \text{sign}(\dot{q}_1) + F_{c1} \dot{q}_1 \\ F_{s2} \text{sign}(\dot{q}_2) + F_{c2} \dot{q}_2 \end{bmatrix} \quad (32)$$

$$J(q) = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix} \quad (33)$$

$$f_{ext} = \begin{bmatrix} 0 \\ -Mg \end{bmatrix} \quad (34)$$

where I_1 and I_2 are the inertias of link 1 and 2, respectively, m_1 and m_2 are the masses of link 1 and 2, respectively, l_1 and l_2 are the lengths of link 1 and 2, respectively, r_1 and r_2 are the positions of center of masses of link 1 and 2, respectively, q_1 and q_2 are the joint angles of link 1 and 2, respectively, F_{s1} and F_{s2} are the stiction coefficients of link 1 and 2, respectively, F_{c1} and F_{c2} are the coulomb friction coefficients of link 1 and 2, respectively, g implies the gravitational acceleration constant, $g = 9.8[m/s^2]$, and

$$\begin{aligned} c_1 &= \cos(q_1) & c_2 &= \cos(q_2) \\ c_{12} &= \cos(q_1 + q_2) & s_1 &= \sin(q_1) \\ s_2 &= \sin(q_2) & s_{12} &= \sin(q_1 + q_2). \end{aligned}$$

For the simulation study, the dynamic parameters are set as $l_1 = l_2 = 0.5[m]$, $m_1 = m_2 = 1.0[kg]$, $r_1 = r_2 = 0.2[m]$, $I_1 = I_2 = 0.05[kg \cdot m^2]$, $F_{s1} = F_{s2} = 0.1$, $F_{c1} = F_{c2} = 0.1$, and $M = 0.1[kg]$. Also these parameters are treated as the unknowns to be identified later.

A. Constructing Momentum Regressor

The parameter vector to be identified through the identification algorithm is defined as following form:

$$\theta \triangleq \begin{bmatrix} I_1 + m_2 l_1^2 \\ I_2 \\ m_1 r_1 + m_2 l_1 + M l_1 \\ m_2 r_2 + M l_2 \\ m_2 r_2 l_1 \\ F_{s1} \\ F_{s2} \\ F_{c1} \\ F_{c2} \end{bmatrix} \in \mathbb{R}^9, \quad (35)$$

where the exact result to be obtained will be

$$\theta^T = [0.3, 0.05, 0.75, 0.25, 0.1, 0.1, 0.1, 0.1, 0.1],$$

if the proposed identification algorithm is effective. As suggested in section II, we need to find two subregressors to construct the momentum regressor by using Eqs. (6) and (8) as follows:

$$W_1 = \begin{bmatrix} 0 & 0 & a_{13} & a_{14} & 0 & a_{16} & 0 & a_{18} & 0 \\ 0 & 0 & 0 & a_{24} & a_{25} & 0 & a_{27} & 0 & a_{29} \end{bmatrix} \quad (36)$$

where

$$\begin{aligned} a_{13} &= g c_1 & a_{14} &= g c_{12} \\ a_{16} &= -\text{sign}(\dot{q}_1) & a_{18} &= -\dot{q}_1 \\ a_{24} &= g c_{12} & a_{25} &= -s_2(\dot{q}_1 + \dot{q}_2)\dot{q}_1 \\ a_{27} &= -\text{sign}(\dot{q}_2) & a_{29} &= -\dot{q}_2 \end{aligned}$$

and

$$W_2 = \begin{bmatrix} b_{11} & b_{12} & 0 & 0 & b_{15} & 0 & 0 & 0 & 0 \\ 0 & b_{22} & 0 & 0 & b_{25} & 0 & 0 & 0 & 0 \end{bmatrix} \quad (37)$$

where

$$\begin{aligned} b_{11} &= \dot{q}_1 & b_{12} &= \dot{q}_1 + \dot{q}_2 \\ b_{15} &= c_2(2\dot{q}_1 + \dot{q}_2) & b_{22} &= \dot{q}_1 + \dot{q}_2 \\ b_{25} &= c_2\dot{q}_1. \end{aligned}$$

Now we can get the momentum regressor as either continuous-time form of Eq. (11) or discrete-time form of Eq. (13). To simplify the simulation and the related discussions, the control input torques are set as follows:

$$\tau[i] = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{for } i = 1, 2, \dots$$

For the simulation study, the Eqs. (13) and (14) will be utilized in the following subsection.

B. Identification Result of the Proposed Scheme

The pseudo code of the proposed identification scheme in the section III is summarized as follows:

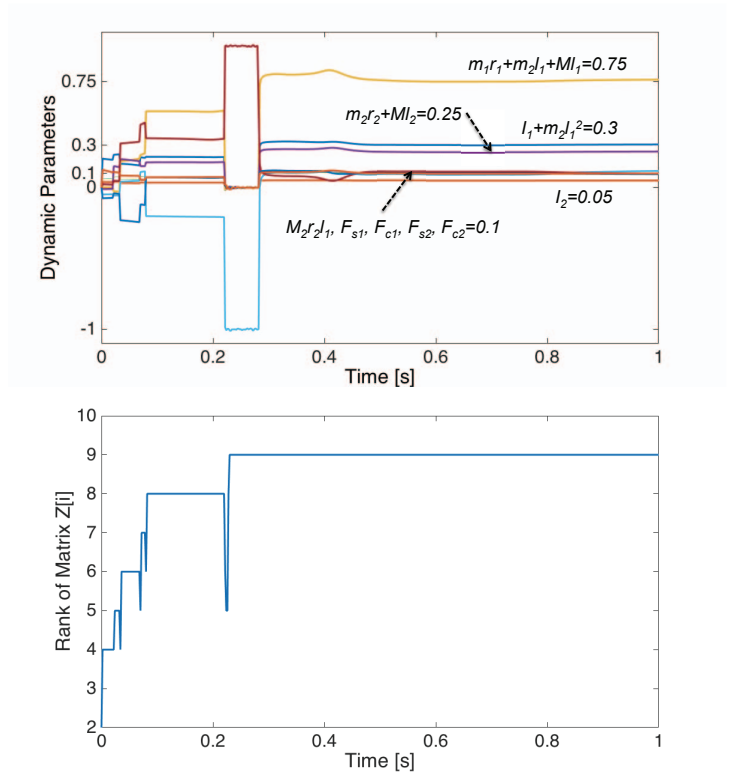


Fig. 3: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0 \cdot I$

```

 $\Omega[0] = 0 \cdot I$       (note that  $\Omega[i] \in \mathbb{R}^{9 \times 9}$ )
 $\Gamma[0] = 0$         (note that  $\Gamma[i] \in \mathbb{R}^9$ )
 $\delta t = 2[ms]$ 
 $n = \frac{t_f}{\delta t}$       ( $t_f$  is a terminal time of simulation)
for  $i = 1 : n$ 
    time  $t = i \cdot \delta t$ 
    generate  $\tau[t] = 1$       (note that  $\tau[i] \in \mathbb{R}^2$ )
    solve Eq. (1):  $q[t], \dot{q}[t]$ 
    construct momentum regressor Eq. (12):
         $Y_H[i], u[i]$  (note that  $Y_H[i] \in \mathbb{R}^{2 \times 9}$  and  $u[i] \in \mathbb{R}^2$ )
         $\Omega[i] = \Omega[i-1] + Y_H^T[i] Y_H[i]$ 
         $\Gamma[i] = \Gamma[i-1] + Y_H^T[i] u[i]$ 
         $W[i] = (I + \Omega^T[i] \Omega[i])^{-1}$  (note that  $W[i] \in \mathbb{R}^{9 \times 9}$ )
         $Z[i] = (\Omega[i] W[i] \Omega^T[i])^+$  (note that  $Z[i] \in \mathbb{R}^{9 \times 9}$ )
         $\Omega_W^+[i] = W[i] \Omega[i]^T Z[i]$ 
         $\hat{\theta}[i] = \Omega_W^+[i] \Gamma[i] + (I - \Omega_W^+[i] \Omega[i]) W[i] \hat{\theta}[i-1]$ 
end

```

To begin with, the identification result obtained by using the proposed scheme is shown in Fig. 3(a) when the zero initial condition, namely $\Omega[0] = 0 \cdot I$, was used. Here we can confirm that the proposed identification scheme has

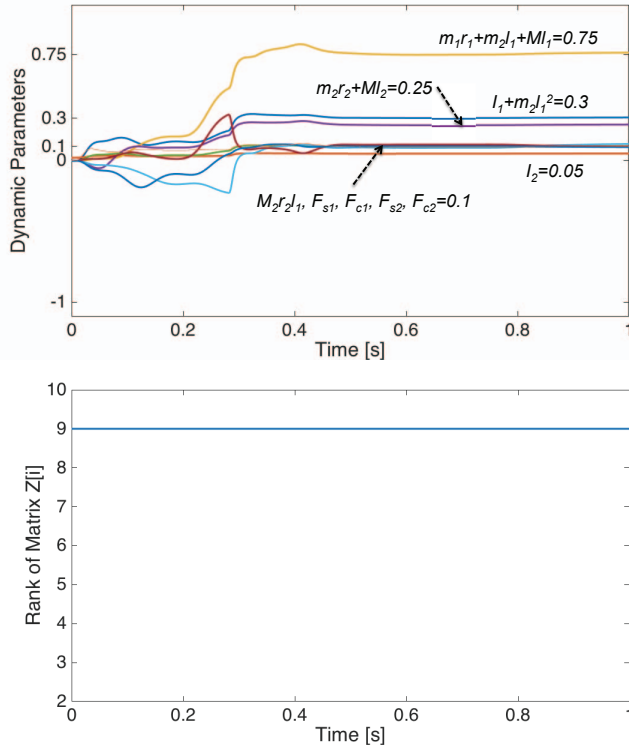


Fig. 4: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0.0001 \cdot I$

produced the exact result after undergoing several discontinuous phenomena. To illustrate the reason of discontinuities, we have suggested the rank variation graph of the matrix $Z[i]$ as shown in Fig. 3(b). Whenever the rank of matrix $Z[i]$ varies, the identification results jump to another values, ultimately until arriving at the exact results. After being satisfied with the full rank condition (here 9) of the matrix $Z[i]$, the identification results are fastly stabilized towards to the exact values.

On the other hand if the initial conditions are assigned to arbitrary small values, such as $\Omega[0] = 0.0001 \cdot I$, then the discontinuity phenomena disappear as shown in Fig. 4. Furthermore, if we increase the initial condition of $\Omega[0]$ to $0.01 \cdot I$, then more stable and smooth results are obtained as shown in Fig. 5 but the convergence speed becomes also slower compared with the identification results obtained from the zero initial condition. These good identification results could be obtained thanks to the momentum regressor and the identification scheme proposed newly in the paper.

V. CONCLUSION

The paper has proposed the momentum regressor as new form and the identification algorithm based on the Lagrange multiplier and optimization. The momentum regressor was suggested through the integration of the conventional Lagrange regressor. It has a robustness property against the measurement noise because it made use of the numerical integration. Finally the simulations were conducted to show the effectiveness of the proposed method.

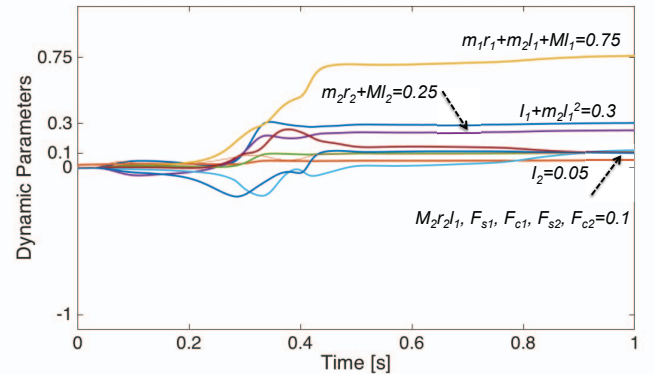


Fig. 5: Simulation result obtained by using the proposed scheme when the initial $\Omega[0] = 0.01 \cdot I$

For future works, we are to develop an adaptive control based on the proposed identification scheme. In addition, the comparative studies with existing adaptive control schemes are implemented to show the relative superiority of the proposed scheme.

REFERENCES

- [1] T. Söderström and P. Stoica, *System Identification*, Prentice Hall, 1989.
- [2] P. K. Khosla, Categorization of Parameters in the Dynamics Robot Model, *IEEE Trans. on Robotics and Automation*, vol. 5, No. 3, pp. 261-268, 1989.
- [3] C. C. de Wit, B. Sciliano, and G. Bastin, *Theory of Robot Control*, Springer, 1996.
- [4] M. W. Spong, S. Hutchinson and M. Vidyasagar, *Robot Modeling and Control*, Wiley, 2005.
- [5] F. L. Lewis, C. T. Abdallah and D. M. Dawson, *Control of Robot Manipulators*, Macmillan, 1993.
- [6] R. Featherstone, *Rigid Body Dynamics Algorithms*, Springer, 2008.
- [7] J. Naudet, and D. Lefeber, General Formulation of an Efficient Recursive Algorithm Based on Canonical Momenta for Forward Dynamics of Closed-Loop Multibody Systems, *ASME 2005 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, pp. 177-185, 2005.
- [8] S. J. Leon, *Linear Algebra with Application*, 8th Edition, Pearson, 2010.
- [9] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, John Wiley & Sons, 1998.
- [10] Y. Choi and J. Cheong, "New Expressions of 2 x 2 Block Matrix Inversion," *IEEE Transactions on Automatic Control*, vol. 54, No. 11, pp. 2648-2653, 2009.
- [11] B. S. Chen, Y. C. Chang and T. C. Lee, "Adaptive Control in Robotic Systems with $\mathcal{H}_\infty$ Tracking Performance", *Automatica*, vol. 33, No. 2, pp. 227-234, 1997.
- [12] J.-J. E. Slotine and W. Li, Composite Adaptive Control of Robot Manipulators, *Automatica*, vol. 25, No. 4, pp. 509-519, 1989.
- [13] R. M. Rangayyan, *Biomedical Signal Analysis*, IEEE Press, 2002.